Discrete Mathematics and Graph Theory

Department of Electronics and Telecommunication Engineering

Ref: Kenneth Rosen, Discrete Mathematics and its Applications, 7th Edition, McGraw Hill Publications, 2012

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Discrete Mathematics

- Discrete mathematics is a branch of mathematics that deals with countable and distinct objects rather than continuous ones

- It focuses on structures and processes that are fundamentally different from continuous mathematics, which deals with objects that can vary smoothly, such as real numbers.

Applications

Discrete Mathematics has a wide range of applications across several fields such as:

- Computer Science and Applications
- Telecommunications and Networking
- Operations Research and Optimization
- Cryptography and Network Security
- Combinatorial Designs and Coding Theory
- Google Maps
- Railway Planning

Fundamentals of Discrete Mathematics

Module 1

- Propositional logic, Conditional statements, Truth tables of Compound propositions, Logic and Bit Operations
- Propositional equivalences, De Morgan‟s Laws, Propositional satisfiability
- Predicates and Quantifiers, Nested Quantifiers, Translating mathematical statements into statements involving nested quantifiers
- Inference, Rules of inference for propositional logic

Propositional logic

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

EXAMPLE 1 All the following declarative sentences are propositions.

1. Washington, D.C., is the capital of the United States of America.
2. Toronto is the capital of Canada.
3. 1 + 1 = 2.
4. 2 + 2 = 3.

Propositions 1 and 3 are true, whereas 2 and 4 are false.

EXAMPLE 2 Consider the following sentences.

1. What time is it?
2. Read this carefully.
3. x + 1 = 2.
4. x + y = z.

Sentences 1 and 2 are not propositions because they are not declarative sentences. Sentences 3 and 4 are not propositions because they are neither true nor false.

Propositional logic

- We use letters to denote propositional variables (or statement variables), that is, variables that represent propositions, just as letters are used to denote numerical variables.
- The conventional letters used for propositional variables are p, q, r, s, ...
- The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.
- The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**.
- Many mathematical statements are constructed by combining one or more propositions. New propositions, called **compound propositions**, are formed from existing propositions using logical operators.

Compound Propositions

Negation

Let p be a proposition. The *negation of* p, denoted by $\neg p$ (also denoted by $\overline{p}$), is the statement

"It is not the case that p."

The proposition $\neg p$ is read "not p." The truth value of the negation of p, $\neg p$, is the opposite of the truth value of p.

Find the negation of the proposition "Vandana's smartphone has at least 32GB of memory" and express this in simple English

This negation can be expressed as "Vandana's smartphone does not have at least 32GB of memory"

Negation

TABLE 1 The Truth Table for the Negation of a Proposition.

p	$\neg p$
T	F
F	T

- The truth table has a row for each of the two possible truth values of a proposition p.
- Each row shows the truth value of $\sim p$ corresponding to the truth value of p for this row.
- The negation of a proposition can also be considered the result of the operation of the negation operator on a proposition.
- The negation operator constructs a new proposition from a single existing proposition.

Conjunction

Let p and q be propositions. The *conjunction* of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Find the conjunction of the propositions p and q where p is the proposition "Tom's PC has more than 16 GB free hard disk space" and q is the proposition "The processor in Tom's PC runs faster than 1 GHz.

Conjunction: "Tom's PC has more than 16 GB free hard disk space, and its processor runs faster than 1 GHz."

For this conjunction to be true, both conditions given must be true. It is false, when one or both of these conditions are false.

Disjunction (Inclusive OR)

Let p and q be propositions. The *disjunction* of p and q, denoted by $p \vee q$, is the proposition "p or q." The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Find the conjunction of the propositions p and q where p is the proposition "Students who have taken calculus can take this class" and q is the proposition "Students who have taken computer science can take this class" .

"Students who have taken calculus or computer science can take this class."

Inclusive OR: A disjunction is true when at least one of the two propositions is true.

Truth Tables of Conjunction and Disjunction

TABLE 2 The Truth Table for the Conjunction of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

TABLE 3 The Truth Table for the Disjunction of Two Propositions.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Disjunction (Exclusive OR)

Let p and q be propositions. The *exclusive or* of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

Exclusive OR: "Students who have taken calculus or computer science, but not both, can enroll in this class."

TABLE 4 The Truth Table for the Exclusive Or of Two Propositions.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statements

Let p and q be propositions. The *conditional statement* $p \rightarrow q$ is the proposition "if p, then q." The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the *hypothesis* (or *antecedent* or *premise*) and q is called the *conclusion* (or *consequence*).

The statement p → q is called a conditional statement because p → q asserts that q is true on the condition that p holds. A conditional statement is also called an **implication**.

TABLE 5 The Truth Table for the Conditional Statement $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

General ways to Express Conditional Statements

"if p, then q"
"if p, q"
"p is sufficient for q"
"q if p"
"q when p"
"a necessary condition for p is q"
"q unless ~p"

"p implies q"
"p only if q"
"a sufficient condition for q is p"
"q whenever p"
"q is necessary for p"
"q follows from p"

Example

- The pledge many politicians make when running for office is "If I am elected, then I will lower taxes."
- "If you get 100% on the final, then you will get an A."

Examples

Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job." Express the statement p → q as a statement in English.

- "If Maria learns discrete mathematics, then she will find a good job."
- "Maria will find a good job when she learns discrete mathematics."
- "For Maria to get a good job, it is sufficient for her to learn discrete mathematics."
- "Maria will find a good job unless she does not learn discrete mathematics."

First statement is true unless Maria learns discrete mathematics, but she does not get a good job,

Examples

“If John has a smartphone, then 2 + 3 = 5” is true from the definition of a conditional statement, because its conclusion is true.
The truth value of the hypothesis does not matter then

“If John has a smartphone, then 2 + 3 = 6” is true if John does not have a smartphone, even though 2 + 3 = 6 is false

We would not use these last two conditional statements in natural language (except perhaps in sarcasm)

The mathematical concept of a conditional statement is independent of a cause- and effect relationship between hypothesis and conclusion.

Examples

What is the value of the variable x after the statement if 2 + 2 = 4 then x := x + 1. if x = 0 before this statement is encountered? (The symbol := stands for assignment. The statement x := x + 1 means the assignment of the value of x + 1 to x.)

Solution: Because 2 + 2 = 4 is true, the assignment statement x := x + 1 is executed. Hence, x has the value 0 + 1 = 1 after this statement is encountered.

Converse, Contrapositive, and Inverse

- We can form some new conditional statements starting with a conditional statement $p \rightarrow q$.
- There are three related conditional statements that occur so often that they have special names.
  - The proposition $q \rightarrow p$ is called the converse of $p \rightarrow q$.
  - The contrapositive of $p \rightarrow q$ is the proposition $\neg q \rightarrow \neg p$.
  - The proposition $\neg p \rightarrow \neg q$ is called the inverse of $p \rightarrow q$.

Example: What are the contrapositive, the converse, and the inverse of the conditional statement "The home team wins whenever it is raining?"

The original statement can be rewritten as "If it is raining, then the home team wins."
Contrapositive: "If the home team does not win, then it is not raining."
Converse: "If the home team wins, then it is raining."
Inverse: "If it is not raining, then the home team does not win."

Equivalent Propositions

- When two compound propositions always have the same truth value we call them equivalent,
- A conditional statement and its contrapositive are equivalent.
- The converse and the inverse of a conditional statement are also equivalent, but neither is equivalent to the original conditional statement.

Biconditionals

Let p and q be propositions. The *biconditional statement* $p \leftrightarrow q$ is the proposition "p if and only if q." The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called *bi-implications*.

EXAMPLE Let p be the statement "You can take the flight," and let q be the statement "You buy a ticket."

$p \leftrightarrow q$ is the statement
"You can take the flight if and only if you buy a ticket."

TABLE 6 The Truth Table for the Biconditional $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Precedence of Logical Operators and Truth Tables of Compound Propositions

TABLE 8
Precedence of Logical Operators.

Operator	*Precedence*
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$.

TABLE 7 The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Logic and Bit Operations

- Computers represent information using bits.A bit is a symbol with two possible values, namely, 0 (zero) and 1 (one).
- This meaning of the word bit comes from binary digit, because zeros and
- ones are the digits used in binary representations of numbers.
- A bit can be used to represent a truth value, because there are two truth values, namely, true and false.
- We use a 1 bit to represent true and a 0 bit to represent false.
- A variable is called a Boolean variable if its value is either true or false. Consequently, a Boolean variable can be represented using a bit.

Truth Tables of Logic and Bit Operations

Truth Value	*Bit*
T	1
F	0

TABLE 9 Table for the Bit Operators *OR*, *AND*, and *XOR*.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Strings

A *bit string* is a sequence of zero or more bits. The *length* of this string is the number of bits in the string.

For eg. 101010011 is a bit string of length nine.

EXAMPLE Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit strings 01 1011 0110 and 11 0001 1101.

Solution: The bitwise *OR*, bitwise *AND*, and bitwise *XOR* of these strings are obtained by taking the *OR*, *AND*, and *XOR* of the corresponding bits, respectively.

This gives us

11 1011 1111 bitwise *OR*
01 0001 0100 bitwise *AND*
10 1010 1011 bitwise *XOR*

Applications of Propositional Logic

Translating English Sentences

How can this English sentence be translated into a logical expression?
“You can access the Internet from campus only if you are a computer science major or you are not a freshman.

a is the proposition “You can access the Internet from campus,”
b is the proposition “You are a computer science major,”
c is the proposition “You are a freshman,” respectively.
Noting that “only if” is one way a conditional statement can be expressed, this sentence can be represented as $a \rightarrow (c \vee \neg f)$.

Applications of Propositional Logic

Express the specification "The automated reply cannot be sent when the file system is full" using logical connectives.

let p denote "The automated reply can be sent" and q denote "The file system is full."

Specification can be represented by the conditional statement $q \rightarrow \neg p$.

Smullyan posed many puzzles about an island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B. What are A and B if A says "B is a knight" and B says "The two of us are opposite types

Applications of Propositional Logic

Logic Circuits

A logic circuit (or digital circuit) receives input signals p1, p2, . . . , pn, each a bit [either 0 (off) or 1 (on)], and produces output signals s1, s2, . . . , sn, each a bit.

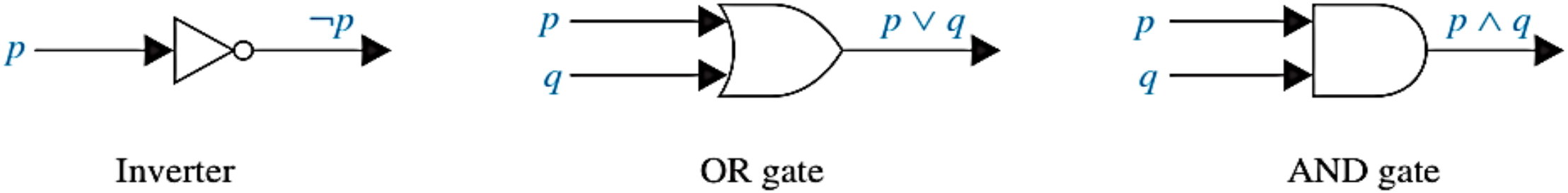
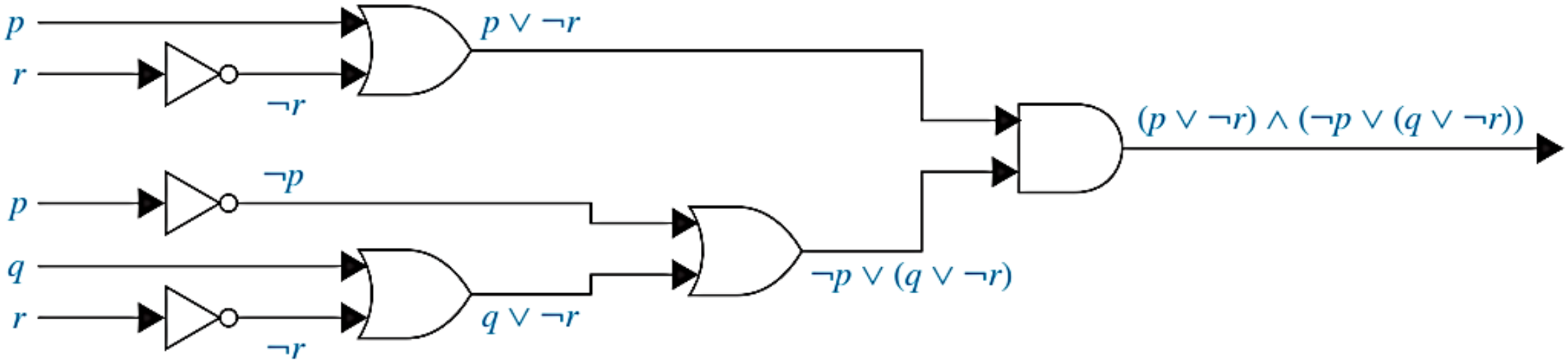


FIGURE 1 **Basic logic gates.**

Applications of Propositional Logic

Build a digital circuit that produces the output (p ∨ ¬r) ∧ (¬p ∨ (q ∨ ¬r)) when given input bits p, q, and r.



Compounded Propositions

"Compound proposition" to refer to an expression formed from propositional variables using logical operators

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a *tautology*. A compound proposition that is always false is called a *contradiction*. A compound proposition that is neither a tautology nor a contradiction is called a *contingency*.

TABLE 1 Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logical Equivalence

The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

- The symbol ≡ is not a logical connective, and p ≡ q is not a compound proposition but rather is the statement that p ↔ q is a tautology.
- The symbol ⇔ is sometimes used instead of ≡ to denote logical equivalence.
- One way to determine whether two compound propositions are equivalent is to use a truth table.

De Morgan's Laws

TABLE 2 De Morgan's Laws.

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

TABLE 3 Truth Tables for $\neg(p \vee q)$ and $\neg p \wedge \neg q$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Example of Logical Equivalence

Show that p ∨ (q ∧ r) and (p ∨ q) ∧ (p ∨ r) are logically equivalent. This is the distributive law of disjunction over conjunction.

TABLE 5 A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Logical Equivalances

TABLE 6 Logical Equivalences.

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Logical Equivalances

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$p \to q \equiv \neg p \vee q$$

$$p \to q \equiv \neg q \to \neg p$$

$$p \vee q \equiv \neg p \to q$$

$$p \wedge q \equiv \neg(p \to \neg q)$$

$$\neg(p \to q) \equiv p \wedge \neg q$$

$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$

$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$

$$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$

$$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Constructing New Logical Equivalences

Show that ¬(p → q) and p ∧ ¬q are logically equivalent.

$$\begin{aligned} \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) \\ &\equiv \neg(\neg p) \wedge \neg q \\ &\equiv p \wedge \neg q \end{aligned}$$

Implications relation to negation and disjunction
by the second De Morgan law
by the double negation law

Constructing New Logical Equivalences

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Solution: We will use one of the equivalences in Table 6 at a time, starting with $\neg(p \vee (\neg p \wedge q))$ and ending with $\neg p \wedge \neg q$. (*Note:* we could also easily establish this equivalence using a truth table.) We have the following equivalences.

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the second De Morgan law}\\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the first De Morgan law}\\
&\equiv \neg p \wedge (p \vee \neg q) && \text{by the double negation law}\\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the second distributive law}\\
&\equiv \mathbf{F} \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv \mathbf{F}\\
&\equiv (\neg p \wedge \neg q) \vee \mathbf{F} && \text{by the commutative law for disjunction}\\
&\equiv \neg p \wedge \neg q && \text{by the identity law for } \mathbf{F}
\end{aligned}
$$

Consequently $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

Constructing New Logical Equivalences

Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Solution: To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to **T**. (*Note:* This could also be done using a truth table.)

$$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$$

BY implication relation with conjunction and disjunction

$$\equiv (\neg p \vee \neg q) \vee (p \vee q)$$

by the first De Morgan law

$$\equiv (\neg p \vee p) \vee (\neg q \vee q)$$

by the associative and commutative laws for disjunction

$$\equiv \mathbf{T} \vee \mathbf{T}$$

Commutative law for disjunction

$$\equiv \mathbf{T}$$

by the domination law

Propositional Satisfiability

- A compound proposition is satisfiable if there is an assignment of truth values to its variables that makes it true.
- When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is unsatisfiable.
- Note that a compound proposition is unsatisfiable if and only if its negation is true for all assignments of truth values to the variables, that is, if and only if its negation is a tautology.

Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable

The problem can be solved using the truth table.

Applications of Satisfiability

Many problems, in diverse areas such as robotics, software testing, computer-aided design, machine vision, integrated circuit design, computer networking, and genetics, can be modeled in terms of propositional satisfiability.

SUDOKU A Sudoku puzzle is represented by a 9 × 9 grid made up of nine 3 × 3 subgrids, known as blocks, as shown in Figure 1. For each puzzle, some of the 81 cells, called givens, are assigned one of the numbers 1, 2, . . . , 9, and the other cells are blank. The puzzle is solved by assigning a number to each blank cell so that every row, every column, and every one of the nine 3 × 3 blocks contains each of the nine possible numbers. Note that instead of using a 9 × 9 grid, Sudoku puzzles can be based on $n^2 \times n^2$ grids, for any positive integer n, with the $n^2 \times n^2$grid made up of n^2n × n subgrids.

Applications of Satisfiability

- For each cell with a given value, we assert $p(i, j, n)$ when the cell in row i and column j has the given value n.
- We assert that every row contains every number:

$$\bigwedge_{i=1}^{9} \bigwedge_{n=1}^{9} \bigvee_{j=1}^{9} p(i, j, n)$$

- We assert that every column contains every number:

$$\bigwedge_{j=1}^{9} \bigwedge_{n=1}^{9} \bigvee_{i=1}^{9} p(i, j, n)$$

Applications of Satisfiability

- We assert that each of the nine 3×3 blocks contains every number:

$$\bigwedge_{r=0}^{2} \bigwedge_{s=0}^{2} \bigwedge_{n=1}^{9} \bigvee_{i=1}^{3} \bigvee_{j=1}^{3} p(3r+i, 3s+j, n)$$

- To assert that no cell contains more than one number, we take the conjunction over all values of n, n', i, and j where each variable ranges from 1 to 9 and $n \neq n'$ of $p(i, j, n) \rightarrow \neg p(i, j, n')$.

Predicates and Quantifiers

- Statements involving variables, such as “x > 3,” “x = y + 3,” “x + y = z,” and “computer x is under attack by an intruder,” are often found in mathematical assertions, in computer programs, and in system specifications.
- These statements are neither true nor false when the values of the variables are not specified.
- The statement “x is greater than 3” has two parts. The first part, the variable x, is the subject of the statement.
- The second part—the predicate, “is greater than 3”—refers to a property that the subject of the statement can have.
- We can denote the statement “x is greater than 3” by P (x), where P denotes the predicate “is greater than 3” and x is the variable.
- The statement P (x) is also said to be the value of the propositional function P at x.
- Once a value has been assigned to the variable x, the statement P (x) becomes a proposition and has a truth value

Examples of Predicates

1. Let P (x) denote the statement "x > 3." What are the truth values of P (4) and P (2)?
Solution: We obtain the statement P (4) by setting x = 4 in the statement "x > 3." Hence, P (4), which is the statement "4 > 3," is true. However, P (2), which is the statement "2 > 3," is false.

Let Q(x, y) denote the statement "x = y + 3." What are the truth values of the propositions Q(1, 2) and Q(3, 0)?
Solution: To obtain Q(1, 2), set x = 1 and y = 2 in the statement Q(x, y). Hence, Q(1, 2) is the statement "1 = 2 + 3," which is false. The statement Q(3, 0) is the proposition "3 = 0 + 3," which is true.

Examples of Predicates

2. Let A(x) denote the statement "Computer x is under attack by an intruder." Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are truth values of A(CS1), A(CS2), and A(MATH1)?

Solution: We obtain the statement A(CS1) by setting x = CS1 in the statement "Computer x is under attack by an intruder." Because CS1 is not on the list of computers currently under attack, we conclude that A(CS1) is false. Similarly, because CS2 and MATH1 are on the list of computers under attack, we know that A(CS2) and A(MATH1) are true.

Examples of Predicates

What are the truth values of the propositions R(1, 2, 3) and R(0, 0, 1)?
Solution: The proposition R(1, 2, 3) is obtained by setting x = 1, y = 2, and z = 3 in the statement R(x, y, z). We see that R(1, 2, 3) is the statement "1 + 2 = 3," which is true. Also note that R(0, 0, 1), which is the statement "0 + 0 = 1," is false.

Consider the statement
if x > 0 then x := x + 1.
When this statement is encountered in a program, the value of the variable x at that point in the execution of the program is inserted into P (x), which is "x > 0." If P (x) is true for this value of x, the assignment statement x := x + 1 is executed, so the value of x is increased by 1. If P (x) is false for this value of x, the assignment statement is not executed, so the value of x is not changed.

Preconditions and Postconditions

- Predicates are also used to establish the correctness of computer programs, that is, to show that computer programs always produce the desired output when given valid input.
- The statements that describe valid input are known as preconditions and the conditions that the output should satisfy when the program has run are known as postconditions.

Consider the following program, designed to interchange the values of two variables x and y.

temp := x x := y y := temp

- For the precondition, we need to express that x and y have particular values before we run the program. So, for this precondition we can use the predicate P (x, y), where P (x, y) is the statement "x = a and y = b," where a and b are the values of x and y before we run the program.
- Because we want to verify that the program swaps the values of x and y for all input values, for the postcondition we can use Q(x, y), where Q(x, y) is the statement "x = b and y = a."

Quantifiers

Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words all, some, many, none, and few are used in quantifications.

THE UNIVERSAL QUANTIFIER

- Many mathematical statements assert that a property is true for all values of a variable in a particular domain, called the domain of discourse (or the universe of discourse), often just referred to as the domain. Such a statement is expressed using universal quantification.
- The universal quantification of P (x) for a particular domain is the proposition that asserts that P (x) is true for all values of x in this domain.

Quantifiers

The *universal quantification* of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain."

The notation $\forall x\, P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier.** We read $\forall x\, P(x)$ as "for all $x\, P(x)$" or "for every $x\, P(x)$." An element for which $P(x)$ is false is called a **counterexample** of $\forall x\, P(x)$.

TABLE 1 Quantifiers.

Statement	*When True?*	*When False?*
$\forall x\, P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x\, P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

Examples of Quantifiers

1. Let P (x) be the statement "x + 1 > x." What is the truth value of the quantification ∀xP (x), where the domain consists of all real numbers?
Solution: Because P (x) is true for all real numbers x, the quantification ∀xP (x) is true.

2. Let Q(x) be the statement "x < 2." What is the truth value of the quantification ∀xQ(x), where the domain consists of all real numbers?
Solution: Q(x) is not true for every real number x, because, for instance, Q(3) is false. That is, x = 3 is a counterexample for the statement ∀xQ(x). Thus ∀xQ(x) is false

3. What is the truth value of ∀xP (x), where P (x) is the statement "x2 < 10" and the domain consists of the positive integers not exceeding 4?
Solution: The statement ∀xP (x) is the same as the conjunction P (1) ∧ P (2) ∧ P (3) ∧ P (4), because the domain consists of the integers 1, 2, 3, and 4. Because P (4), which is the statement "42 < 10," is false, it follows that ∀xP (x) is false.

Existential Quantifier

The *existential quantification* of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$."

We use the notation $\exists x\, P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

Let P (x) denote the statement "x > 3." What is the truth value of the quantification ∃xP (x), where the domain consists of all real numbers?

Solution: Because "x > 3" is sometimes true—for instance, when x = 4—the existential quantification of P(x), which is ∃xP (x), is true. The statement ∃xP (x) is false if and only if there is no element x in the domain for which P (x) is true. That is, ∃xP (x) is false if and only if P (x) is false for every element of the domain.

Let Q(x)denote the statement "x = x + 1."What is the truth value of the quantification ∃xQ(x), where the domain consists of all real numbers?

Solution: Because Q(x) is false for every real number x, the existential quantification of Q(x), which is ∃xQ(x), is false

Uniqueness Quantifier

- Uniqueness quantifier, denoted by ∃! or $\exists_1$. The notation ∃!xP (x) [or $\exists_1$xP (x)] states “There exists a unique x such that P (x) is true.”
- Other phrases for uniqueness quantification include “there is exactly one” and “there is one and only one.”
- For instance, ∃!x(x - 1 = 0), where the domain is the set of real numbers, states that there is a unique real number x such that x - 1 = 0. This is a true statement, as x = 1 is the unique real number such that x - 1 = 0.

Quantifiers with Restricted Domains

What do the statements $\forall x < 0\ (x^2 > 0)$, $\forall y = 0\ (y^3 = 0)$, and $\exists z > 0\ (z^2 = 2)$ mean, where the domain in each case consists of the real numbers?

Solution: The statement $\forall x < 0\ (x^2 > 0)$ states that for every real number x with $x < 0$, $x^2 > 0$. That is, it states "The square of a negative real number is positive." This statement is the same as $\forall x(x < 0 \rightarrow x^2 > 0)$.

The statement $\forall y = 0\ (y^3 = 0)$ states that for every real number y with $y = 0$, we have $y^3 = 0$. That is, it states "The cube of every nonzero real number is nonzero." Note that this statement is equivalent to $\forall y(y = 0 \rightarrow y^3 = 0)$.

Finally, the statement $\exists z > 0\ (z^2 = 2)$ states that there exists a real number z with $z > 0$ such that $z^2 = 2$. That is, it states "There is a positive square root of 2." This statement is equivalent to $\exists z(z > 0 \wedge z^2 = 2)$.

Precedence of Quantifiers

- The quantifiers ∀ and ∃ have higher precedence than all logical operators from propositional calculus. For example, ∀xP (x) ∨ Q(x) is the disjunction of ∀xP (x) and Q(x).
- In other words, it means (∀xP (x)) ∨ Q(x) rather than ∀x(P (x) ∨ Q(x))

Logical Equivalences Involving Quantifiers

- Statements involving predicates and quantifiers are logically equivalent if and only if they have the same truth value no matter which predicates are substituted into these statements and which domain of discourse is used for the variables in these propositional functions.
- We use the notation S ≡ T to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.

Negating Quantified Expressions

Consider the negation of the statement
“Every student in your class has taken a course in calculus.” This statement is a universal quantification, namely, $\forall x P(x)$,

- The negation of this statement is “It is not the case that every student in your class has taken a course in calculus.”
- This is equivalent to “There is a student in your class who has not taken a course in calculus.”
- And this is simply the existential quantification of the negation of the original propositional function, namely, $\exists x \neg P(x)$.

De Morgan's Laws for Quantifiers

TABLE 2 De Morgan's Laws for Quantifiers.

Negation	*Equivalent Statement*	*When Is Negation True?*	*When False?*
$\neg\exists x\, P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall x\, P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Solution: The negation of $\forall x(x^2 > x)$ is the statement $\neg\forall x(x^2 > x)$, which is equivalent to $\exists x\neg(x^2 > x)$. This can be rewritten as $\exists x(x^2 \leq x)$. The negation of $\exists x(x^2 = 2)$ is the statement $\neg\exists x(x^2 = 2)$, which is equivalent to $\forall x\neg(x^2 = 2)$. This can be rewritten as $\forall x(x^2 = 2)$. The truth values of these statements depend on the domain.

Using Quantifiers in System Specifications

Consider these statements. The first two are called premises and the third is called the conclusion. The entire set is called an argument.
"All lions are fierce."
"Some lions do not drink coffee."
"Some fierce creatures do not drink coffee."

Solution: We can express these statements as: $\forall x(P(x) \rightarrow Q(x))$. $\exists x(P(x) \wedge \neg R(x))$. $\exists x(Q(x) \wedge \neg R(x))$.

Second statement cannot be written as $\exists x(P(x) \rightarrow \neg R(x))$.
The reason is that $P(x) \rightarrow \neg R(x)$ is true whenever x is not a lion, so that $\exists x(P(x) \rightarrow \neg R(x))$ is true as long as there is at least one creature that is not a lion, even if every lion drinks coffee. Similarly, the third statement cannot be written as $\exists x(Q(x) \rightarrow \neg R(x))$.

Nested Quantifiers

$\forall x \forall y(x + y = y + x)$ says that $x + y = y + x$ for all real numbers x and y.
This is the commutative law for addition of real numbers.

Likewise, the statement $\forall x \exists y(x + y = 0)$
says that for every real number x there is a real number y such that $x + y = 0$.
This states that every real number has an additive inverse.

Similarly, the statement $\forall x \forall y \forall z(x + (y + z) = (x + y) + z)$ is the associative law for addition of real numbers.

Order of Quantifiers

Let P (x, y) be the statement "x + y = y + x." What are the truth values of the quantifications ∀x∀yP (x, y) and ∀y∀xP (x, y) where the domain for all variables consists of all real numbers?

Solution: The quantification ∀x∀yP (x, y) denotes the proposition
"For all real numbers x, for all real numbers y, x + y = y + x."

Let Q(x, y) denote "x + y = 0." What are the truth values of the quantifications ∃y∀xQ(x, y) and ∀x∃yQ(x, y), where the domain for all variables consists of all real numbers?
Solution: The quantification ∃y∀xQ(x, y) denotes the proposition
"There is a real number y such that for every real number x, Q(x, y)."

Quantifications of Two Variables

TABLE 1 Quantifications of Two Variables.

Statement	*When True?*	*When False?*
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y.	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y.
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y.

Translating Mathematical Statements into Statements Involving Nested Quantifiers

Translate the statement “The sum of two positive integers is always positive” into a logical expression.

Solution: To translate this statement into a logical expression, we first rewrite it so that the implied quantifiers and a domain are shown:

“For every two integers, if these integers are both positive, then the sum of these integers is positive.” Next, we introduce the variables x and y to obtain “For all positive integers x and y, x + y is positive.”

Consequently, we can express this statement as

$\forall x \forall y((x > 0) \wedge (y > 0) \rightarrow (x + y > 0))$, where the domain for both variables consists of all integers.

Translating Mathematical Statements into Statements Involving Nested Quantifiers

Translate the statement “Every real number except zero has a multiplicative inverse.” (A multiplicative inverse of a real number x is a real number y such that xy = 1.)

Solution: We first rewrite this as “For every real number x except zero, x has a multiplicative inverse.” We can rewrite this as “For every real number x, if x = 0, then there exists a real number y such that xy = 1.” This can be rewritten as $\forall x((x = 0) \rightarrow \exists y(xy = 1))$.

Rules of Inference

- Proofs in mathematics are valid arguments that establish the truth of mathematical statements.
- Argument, is a sequence of statements that end with a conclusion.
- Valid means that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises, of the argument.
- That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false.

Consider the following argument involving propositions (which, by definition, is a sequence of propositions): “If you have a current password, then you can log onto the network.” “You have a current password.” Therefore, “You can log onto the network.

Use p to represent “You have a current password” and q to represent “You can log onto the network.” Then, the argument has the form $p \rightarrow q$, p, $\therefore q$
where $\therefore$ is the symbol that denotes “therefore.”

Arguments

An *argument* in propositional logic is a sequence of propositions. All but the final proposition in the argument are called *premises* and the final proposition is called the *conclusion*. An argument is *valid* if the truth of all its premises implies that the conclusion is true.

An *argument form* in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is *valid* no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Rules of Inference for Propositional Logic

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \rightarrow q$ $\therefore q$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\neg q$ $p \rightarrow q$ $\therefore \neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism

Rules of Inference for Propositional Logic

$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \rightarrow (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \rightarrow p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Rules of Inference for Propositional Logic

State which rule of inference is the basis of the following argument: “It is below freezing now. Therefore, it is either below freezing or raining now.”
Solution: Let p be the proposition “It is below freezing now” and q the proposition “It is raining now.”
Then this argument is of the form p , $\therefore p \vee q$
This is an argument that uses the addition rule.

State which rule of inference is the basis of the following argument: “It is below freezing and raining now. Therefore, it is below freezing now.”
Solution: Let p be the proposition “It is below freezing now,” and let q be the proposition “It is raining now.” This argument is of the form $p \wedge q, \therefore p$
This argument uses the simplification rule.

Rules of Inference for Propositional Logic

State which rule of inference is used in the argument:
If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Solution: Let p be the proposition "It is raining today," let q be the proposition "We will not have a barbecue today," and let r be the proposition "We will have a barbecue tomorrow." Then this argument is of the form

$p \rightarrow q, \quad q \rightarrow r, \quad \therefore p \rightarrow r$

Hence, this argument is a hypothetical syllogism

Using Rules of Inference to Build Arguments

Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

Solution: Let p be the proposition "It is sunny this afternoon," q the proposition "It is colder than yesterday," r the proposition "We will go swimming," s the proposition "We will take a canoe trip," and t the proposition "We will be home by sunset."

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Using Rules of Inference to Build Arguments

Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

Solution: Let p be the proposition "You send me an e-mail message," q the proposition "I will finish writing the program," r the proposition "I will go to sleep early," and s the proposition "I will wake up feeling refreshed." Then the premises are p → q, ¬p → r, and r → s.The desired conclusion is ¬q → s.

Step	Reason
1. $p \rightarrow q$	Premise
2. $\neg q \rightarrow \neg p$	Contrapositive of (1)
3. $\neg p \rightarrow r$	Premise
4. $\neg q \rightarrow r$	Hypothetical syllogism using (2) and (3)
5. $r \rightarrow s$	Premise
6. $\neg q \rightarrow s$	Hypothetical syllogism using (4) and (5)

Sample Slide

Tutorial 1

Construct a truth table for each of these compound propositions.

a) $p \rightarrow \neg p$ b) $p \leftrightarrow \neg p$ c) $p \oplus (p \vee q)$ d) $(p \wedge q) \rightarrow (p \vee q)$

Determine whether these biconditionals are true or false.

a) 2 + 2 = 4 if and only if 1 + 1 = 2. b) 1 + 1 = 2 if and only if 2 + 3 = 4.

Find the bitwise OR, bitwise AND, and bitwise XOR of each of these pairs of bit strings.

a) 101 1110, 010 0001 b) 1111 0000, 1010 1010

Explain, using a truth table, why $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when p, q, and r have the same truth value and it is false otherwise.

Prove the following (i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (ii) $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Tutorial 2

1. Determine whether each of these compound propositions is satisfiable.
a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$ b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$ c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

2. Show that each of these conditional statements is a tautology by using truth tables.
a) $(p \wedge q) \rightarrow p$ b) $p \rightarrow (p \vee q)$ c) $\neg p \rightarrow (p \rightarrow q)$ d) $(p \wedge q) \rightarrow (p \rightarrow q)$

3. Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent.

4. Determine the truth value of each of these statements if the domain consists of all real numbers. a) $\exists x(x^3 = -1)$ b) $\exists x(x^4 < x^2)$ c) $\forall x((-x)^2 = x^2)$ d) $\forall x(2x > x)$

5. Let Q(x) be the statement "x + 1 > 2x." If the domain consists of all integers, what are these truth values? a) Q(0) b) Q(−1) c) Q(1) d) $\exists x Q(x)$ e) $\forall x Q(x)$ f) $\exists x \neg Q(x)$

Practice Problems

1. Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow p$, and $\neg s$ and conclusion $q \rightarrow r$ is valid using rules of inference.

2. For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

a) "If I work, it is either sunny or partly sunny." "I worked last Monday or I worked last Friday." "It was not sunny on Tuesday." "It was not partly sunny on Friday."

b) "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragonflies."

3. For each of these arguments determine whether the argument is correct or incorrect and explain why.

a) All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.

b) All parrots like fruit.Mypet bird is not a parrot. Therefore, my pet bird does not like fruit.

4. Determine whether the arguments is valid. If an argument is correct, what rule of inference is being used? If it is not, what logical error occurs? If n is a real number such that n > 1, then $n^2 > 1$. Suppose that $n^2 > 1$. Then n > 1.

5. Let F(x, y) be the statement "x can fool y," where the domain consists of all people in the world. Use quantifiers to express each of these statements. a) There is someone who can fool exactly one person besides himself or herself. b) Everybody can fool somebody. c) There is no one who can fool everybody.

THANK YOU